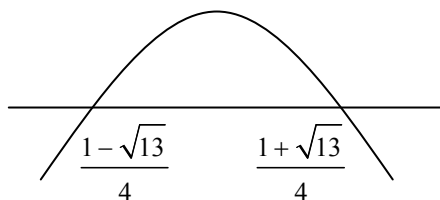


2009 H2 Maths Prelim(Paper 1) Solutions

- 1(i) $(8-x)^{\frac{1}{3}} = 8^{\frac{1}{3}} \left(1 - \frac{x}{8}\right)^{\frac{1}{3}}$
- $$= 2 \left(1 + \frac{1}{3} \left(-\frac{x}{8}\right) + \frac{\frac{1}{3}(\frac{1}{3}-1)}{2!} \left(-\frac{x}{8}\right)^2 + \dots \right)$$
- $$= 2 \left(1 - \frac{1}{24}x - \frac{1}{576}x^2 + \dots \right)$$
- $$= 2 - \frac{1}{12}x - \frac{1}{288}x^2 + \dots$$
- The expansion is valid for $-8 < x < 8$.
- 1(ii) $(8 - (-1))^{\frac{1}{3}} = 2 - \frac{1}{12}(-1) - \frac{1}{288}(-1)^2 + \dots$
- $$(9)^{\frac{1}{3}} \approx \frac{599}{288}$$
- $$\left(\frac{27}{3}\right)^{\frac{1}{3}} \approx \frac{599}{288}$$
- $$\frac{3}{\sqrt[3]{3}} \approx \frac{599}{288}$$
- $$\sqrt[3]{3} \approx \frac{864}{599}$$
- 1(iii) The substitution $x = 5$ will not give a better approximation since $|-1|$ is less than 5.

- 2(i) $\frac{2x+11}{x^2+2} > 4$
- $$\frac{2x+11-4x^2-8}{x^2+2} > 0$$
- Since $x^2+2 > 0$ for all x , we need only to solve $-4x^2+2x+3 > 0$
- When $-4x^2+2x+3 = 0$, $x = \frac{-2 \pm \sqrt{2^2 - 4(-4)(3)}}{2(-4)} = \frac{1 \pm \sqrt{13}}{4}$



Hence, $-4x^2 + 2x + 3 > 0 \Rightarrow \frac{1 - \sqrt{13}}{4} < x < \frac{1 + \sqrt{13}}{4}$

(ii)	$\frac{2x+7}{x^2-4x+6} > 4$ $\frac{2(x-2)+11}{(x-2)^2+2} > 4$ Replace x with $x-2$: $\frac{1-\sqrt{13}}{4} < x-2 < \frac{1+\sqrt{13}}{4}$ $\therefore \frac{9-\sqrt{13}}{4} < x < \frac{9+\sqrt{13}}{4}$
3(i)	$y = e^{3\tan^{-1}x} \Rightarrow \frac{dy}{dx} = \frac{3}{1+x^2} e^{3\tan^{-1}x} = \frac{3y}{1+x^2}$ $\therefore (1+x^2) \frac{dy}{dx} = 3y \text{ (Shown)}$ $(1+x^2) \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} = 3 \frac{dy}{dx} \Rightarrow (1+x^2) \frac{d^2y}{dx^2} + (2x-3) \frac{dy}{dx} = 0$ $(1+x^2) \frac{d^3y}{dx^3} + 2x \frac{d^2y}{dx^2} + (2x-3) \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = 0$ $\Rightarrow (1+x^2) \frac{d^3y}{dx^3} + (4x-3) \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = 0$ When $x=0$, $y=1$, $\frac{dy}{dx}=3$, $\frac{d^2y}{dx^2}=9$, $\frac{d^3y}{dx^3}=21$ Hence, $y = f(0) + xf'(0) + x^2 \frac{f''(0)}{2!} + x^3 \frac{f'''(0)}{3!} + \dots$ $= 1 + 3x + \frac{9}{2}x^2 + \frac{7}{2}x^3 + \dots$
(i)	$e^{2x+3\tan^{-1}x}$ $= e^{2x} e^{3\tan^{-1}x}$ $= \left(1 + 2x + \frac{(2x)^2}{2!} + \dots\right) \left(1 + 3x + \frac{9}{2}x^2 + \frac{7}{2}x^3 + \dots\right)$ $= 1 + 5x + \frac{25}{2}x^2 + \dots$
4(i)	$S_{4n} = \frac{2(1-(-b^2)^{2n})}{1+b^2} + \frac{2b(1-(-b^2)^{2n})}{1+b^2}$ $= \frac{2(1-b^{4n})(1+b)}{1+b^2}$
4(ii)	Series has a sum to infinity when $-b^2 < 1 \Rightarrow b < 1 \Rightarrow -1 < b < 1$

$$S_{\infty} - S_{4n} < 0.001$$

$$\frac{2}{1+b^2} + \frac{2b}{1+b^2} - \frac{2(1-b^{4n})(1+b)}{1+b^2} < 0.001$$

$$\frac{2(1+b)(1-(1-b^{4n}))}{1+b^2} < 0.001$$

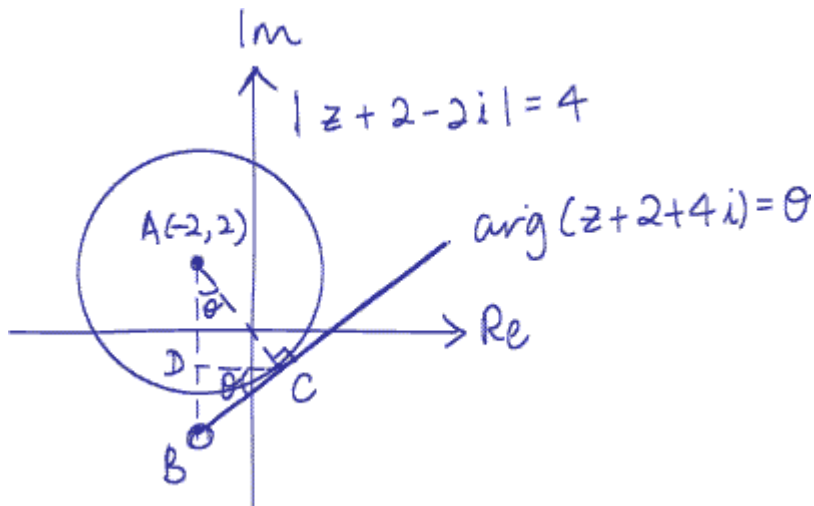
$$\text{Subst. } b = -0.9, (-0.9)^{4n} < 0.00905$$

$$n > \frac{\ln 0.00905}{\ln(-0.9)^4}$$

$$n > 11.2$$

$$\text{Smallest } n = 12$$

5



$$\text{In } \triangle ACB, \cos \theta = \frac{4}{6} = \frac{2}{3}$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta} = \frac{\sqrt{5}}{3} \text{ (Shown)}$$

$$\text{In } \triangle ADC, \sin \theta = \frac{DC}{4}$$

$$DC = 4 \sin \theta = \frac{4}{3} \sqrt{5}$$

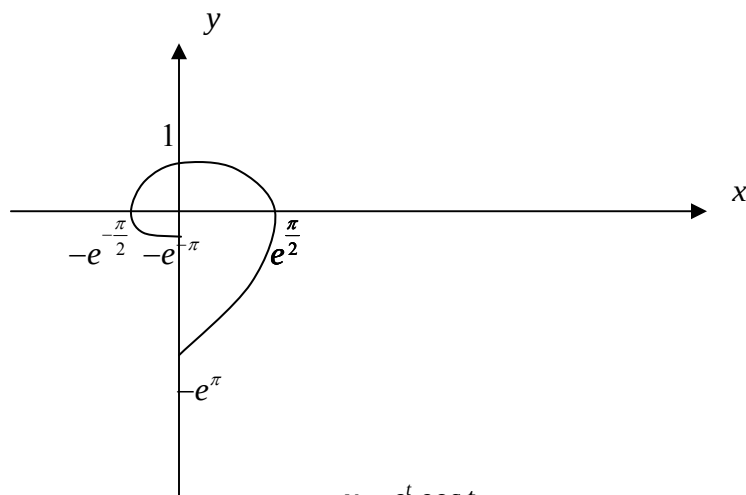
$$AD = \sqrt{4^2 - \left(\frac{4}{3} \sqrt{5}\right)^2} = \frac{8}{3}$$

$$w \equiv \left(\frac{4}{3} \sqrt{5} - 2, -\left(\frac{8}{3} - 2 \right) \right) = \left(\frac{4}{3} \sqrt{5} - 2, -\frac{2}{3} \right)$$

$$w = \frac{4}{3} \sqrt{5} - 2 - \frac{2}{3} i$$

6	Let $\overrightarrow{OA} = \begin{pmatrix} 5 \\ -3 \\ 0 \end{pmatrix}, \quad \overrightarrow{OP} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OB} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$ We have $\overrightarrow{AP} = \begin{pmatrix} -5 \\ 2 \\ 1 \end{pmatrix}, \quad \overrightarrow{AB} = \begin{pmatrix} -4 \\ 2 \\ 1 \end{pmatrix}$ $\overrightarrow{AP} \times \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \\ -9 \end{pmatrix} = -3 \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$ Take normal vector for plane $\pi_1, n_1 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$ $\overrightarrow{AB} \times \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -3 \\ -4 \\ -4 \end{pmatrix} = -1 \begin{pmatrix} 3 \\ 4 \\ 4 \end{pmatrix}$ Take normal vector for plane $\pi_2, n_2 = \begin{pmatrix} 3 \\ 4 \\ 4 \end{pmatrix}$ Hence the direction vector of ℓ_3 has direction vector $\begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} \times \begin{pmatrix} 3 \\ 4 \\ 4 \end{pmatrix} = \begin{pmatrix} -8 \\ 5 \\ 1 \end{pmatrix}$ Vector equation of ℓ_3 is $\mathbf{r} = \begin{pmatrix} 5 \\ -3 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} -8 \\ 5 \\ 1 \end{pmatrix}, \text{ where } \gamma \in \mathbb{R} \quad (\text{Ans})$
6(ii)	For $\pi_3, n_3 = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ Therefore, angle between ℓ_3 and the plane π_3 is given by $\sin \theta = \frac{\left \begin{pmatrix} -8 \\ 5 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right }{\sqrt{90} \sqrt{2}} \Rightarrow \theta = 26.6^\circ \quad (1 \text{ decimal place})$

7(i)



7(ii)

$$x = e^t \sin t$$

$$y = e^t \cos t$$

$$\frac{dx}{dt} = e^t \sin t + e^t \cos t$$

$$\frac{dy}{dt} = e^t \cos t - e^t \sin t$$

$$\frac{dy}{dx} = \frac{e^t (\cos t - \sin t)}{e^t (\sin t + \cos t)}$$

When $t = \frac{\pi}{4}$, $\cos t - \sin t = 0 \Rightarrow$ tangent at $t = \frac{\pi}{4}$ // x-axis.

$$\text{When } t = \frac{\pi}{4}, y = \frac{e^{\frac{\pi}{4}}}{\sqrt{2}}.$$

Hence, the tangent to the curve at $t = \frac{\pi}{4}$ is $y = \frac{e^{\frac{\pi}{4}}}{\sqrt{2}}.$

(iii)

$$P \text{ is } \left(\frac{1}{\sqrt{2}} e^{\frac{\pi}{4}}, \frac{1}{\sqrt{2}} e^{\frac{\pi}{4}} \right) = (1.55, 1.55).$$

$$OP = \sqrt{\frac{1}{2} e^{\frac{\pi}{2}} + \frac{1}{2} e^{\frac{\pi}{2}}} = e^{\frac{\pi}{4}}$$

Given that angle OQ made with the positive x-axis is **bigger** than the angle

OP made with the positive x-axis and $\angle POQ = \frac{\pi}{12}$, $\theta = \frac{\pi}{4} + \frac{\pi}{12} = \frac{\pi}{3}.$

$$\text{In the first quadrant, } \tan \theta = \frac{y}{x} = \frac{e^t \cos t}{e^t \sin t} = \cot t = \tan\left(\frac{\pi}{2} - t\right)$$

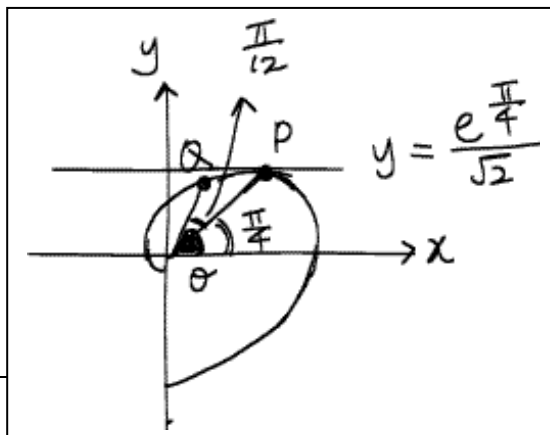
$$\therefore \theta = \frac{\pi}{2} - t$$

$$\text{When } \theta = \frac{\pi}{3}, t = \frac{\pi}{6}$$

Coordinates of Q are :

$$x = e^{\frac{\pi}{6}} \sin \frac{\pi}{6} = \frac{1}{2} e^{\frac{\pi}{6}} \approx 0.844$$

$$y = e^{\frac{\pi}{6}} \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2} e^{\frac{\pi}{6}} \approx 1.462$$



	$OQ = \sqrt{\frac{1}{4}e^{\frac{\pi}{3}} + \frac{3}{4}e^{\frac{\pi}{3}}} = e^{\frac{\pi}{6}}$ $\text{Area of } \triangle POQ = \frac{1}{2}(OQ)(OP)\sin(\angle POQ) = \frac{1}{2}\left(e^{\frac{\pi}{4}}\right)\left(e^{\frac{\pi}{6}}\right)\left(\sin\left(\frac{\pi}{12}\right)\right)$ $= 0.479 \text{ units}^2 \text{ (3 sig. fig)}$
8(a)	$\int \frac{2x^2 - 5x + 13}{x^2 - 2x + 5} dx$ $= \int \left(2 - \frac{x-3}{x^2 - 2x + 5} \right) dx$ $= \int 2 + \frac{-0.5(2x-2)+2}{x^2 - 2x + 5} dx$ $A = 2, B = -0.5, C = 2$ $= \int 2 dx - \frac{1}{2} \int \frac{2x-2}{x^2 - 2x + 5} dx + 2 \int \frac{1}{x^2 - 2x + 5} dx$ $= \int 2 dx - \frac{1}{2} \int \frac{2x-2}{x^2 - 2x + 5} dx + 2 \int \frac{1}{(x-1)^2 + 2^2} dx$ $= 2x - \frac{1}{2} \ln x^2 - 2x + 5 + 2 \left(\frac{1}{2} \right) \tan^{-1} \left(\frac{x-1}{2} \right) + c$
8(b)	Using integration by parts, $\int_1^{2e} x^{n-1} \ln x dx$ $= \left[\frac{x^n \ln x}{n} \right]_1^{2e} - \int_1^{2e} \left(\frac{x^n}{n} \right) \left(\frac{1}{x} \right) dx$ $= \frac{(2e)^n \ln(2e)}{n} - \frac{1}{n} \int_1^{2e} x^{n-1} dx$ $= \frac{(2e)^n (\ln 2 + 1)}{n} - \frac{1}{n} \left[\frac{x^n}{n} \right]_1^{2e}$ $= \frac{(2e)^n (\ln 2 + 1)}{n} - \frac{1}{n} \left[\frac{(2e)^n}{n} - \frac{1}{n} \right]$ $= \frac{1}{n^2} \left[n(2e)^n (\ln 2 + 1) - (2e)^n + 1 \right]$
9 (a)	$x^2 + y^2 = 4 \Rightarrow y = \pm \sqrt{4 - x^2}$ $\frac{x^2}{4} + \frac{y^2}{9} = 1 \Rightarrow y = \pm 3 \sqrt{1 - \frac{x^2}{4}} = \pm \frac{3}{2} \sqrt{4 - x^2}$

	$x = 2 \sin \theta \Rightarrow \frac{dx}{d\theta} = 2 \cos \theta$ When $x = 2$, $\theta = \sin^{-1} 1 = \frac{\pi}{2}$ When $x = -2$, $\theta = \sin^{-1}(-1) = -\frac{\pi}{2}$ Area of region R $= 2 \int_{-2}^2 \left(\frac{3}{2} \sqrt{4-x^2} - \sqrt{4-x^2} \right) dx$ $= \int_{-2}^2 \sqrt{4-x^2} dx$ $= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{4-4\sin^2 \theta} 2\cos \theta d\theta$ $= 4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 \theta d\theta$ $= 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\cos 2\theta + 1) d\theta$ $= 2 \left[\frac{\sin 2\theta}{2} + \theta \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$ $= 2\pi$
9 (b)	$\text{Volume} = 2 \left[\pi \int_0^3 \left(\frac{2}{3} \sqrt{9-y^2} \right)^2 dy \right] - \frac{4}{3} \pi (2)^3$ $= \frac{8}{9} \pi \int_0^3 (9-y^2) dy - \frac{32}{3} \pi$ $= \frac{8}{9} \pi \left[9y - \frac{y^3}{3} \right]_0^3 - \frac{32}{3} \pi$ $= 16\pi - \frac{32}{3} \pi = \frac{16}{3} \pi$
10(i)	Horizontal asymptote : $y = 2$; Vertical asymptote : $x = 0$; $x = -\lambda$. When $y = 0$, $2x^2 + 3x - 15 = 0 \Rightarrow x = -3.59$ or 2.09
(ii)	

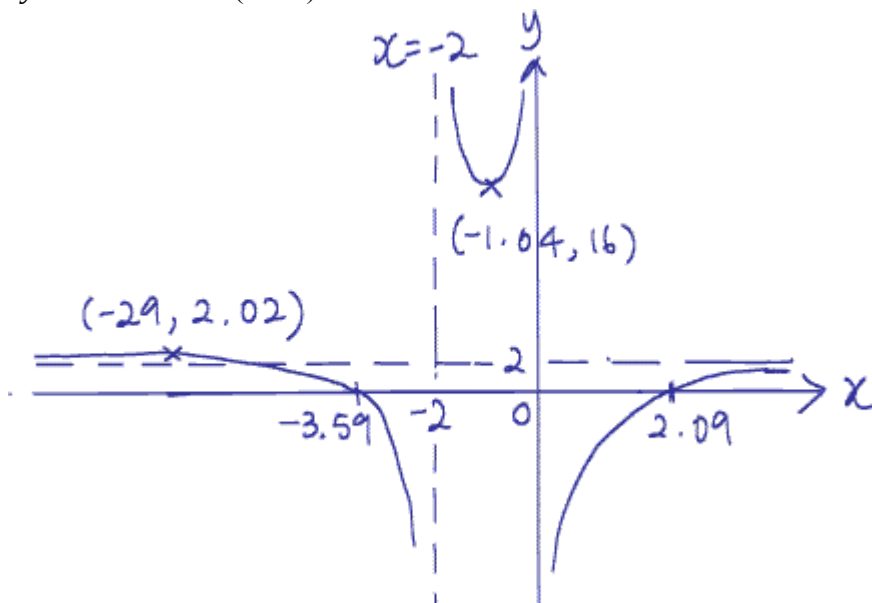
$$y = \frac{2x^2 + 3x - 15}{x(x+2)}$$

$$\frac{dy}{dx} = \frac{(x^2 + 2x)(4x + 3) - (2x^2 + 3x - 15)(2x + 2)}{x^2(x+2)^2} = 0$$

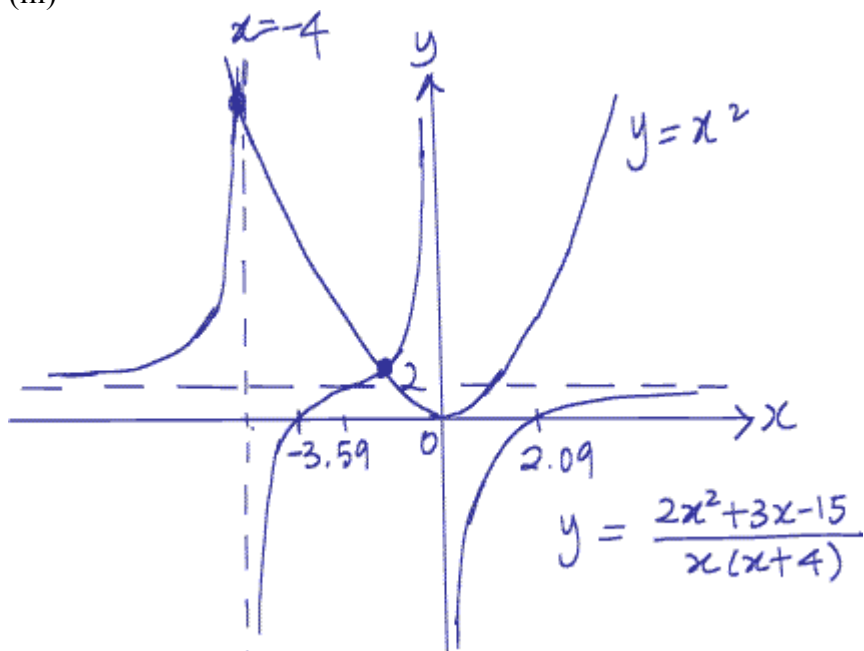
$$x^2 + 30x + 30 = 0$$

$$x = -29.0 \text{ or } -1.04 \text{ (from GC)}$$

$$y = 2.02 \text{ or } 16.0 \text{ (3 s.f.)}$$



(iii)



	$x^4 + 4x^3 - 2x^2 - 3x + 15 = 0$ $x^4 + 4x^3 = 2x^2 + 3x - 15$ $x^2(x^2 + 4x) = 2x^2 + 3x - 15$ $x^2 = \frac{2x^2 + 3x - 15}{x^2 + 4x}$ Another suitable graph is $y = x^2$ The two intersections between the two graphs showed that $x^4 + 4x^3 - 2x^2 - 3x + 15 = 0$ has exactly two real roots.
11 a (i) (ii) (iii)	$z^4 = -64 = 64 e^{i\pi}$ $\Rightarrow z = 2\sqrt[4]{2} e^{i(\frac{\pi}{4} + \frac{2k\pi}{4})} = 2\sqrt[4]{2} e^{i(\frac{\pi}{4} + \frac{k\pi}{2})}, k = -2, -1, 0, 1$ Therefore, $z = 2\sqrt[4]{2} e^{i(\frac{\pi}{4})}, 2\sqrt[4]{2} e^{i(\frac{3\pi}{4})}, 2\sqrt[4]{2} e^{i(-\frac{3\pi}{4})}, 2\sqrt[4]{2} e^{i(-\frac{\pi}{4})}$ $z = 2 + 2i, -2 + 2i, -2 - 2i, 2 - 2i.$ $z^4 + 64 = (z - (2+2i))(z - (-2+2i))(z - (-2-2i))(z - (2-2i))$ $z^4 + 64 = (z^2 - 4z + 8)(z^2 + 4z + 8).$
11 (b)	$\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ $\sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$ Therefore $\cos^2 \theta \sin^4 \theta = \left(\frac{1}{2}(e^{i\theta} + e^{-i\theta})\right)^2 \left(\frac{1}{2i}(e^{i\theta} - e^{-i\theta})\right)^4$ $= \frac{1}{64}[(e^{6i\theta} + e^{-6i\theta}) - 2(e^{4i\theta} + e^{-4i\theta}) - (e^{2i\theta} + e^{-2i\theta}) + 4]$ $= \frac{1}{32}[\cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2] \text{ (Shown)}$ $\int_0^{\pi/3} \cos^2 \theta \sin^4 \theta \, d\theta$ $= \int_0^{\pi/3} \frac{1}{32}[\cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2] \, d\theta$ $= \frac{1}{32} \left[\frac{\sin 6\theta}{6} - 2 \frac{\sin 4\theta}{4} - \frac{\sin 2\theta}{2} + 2\theta \right]_0^{\pi/3}$ $= \frac{\pi}{48}.$

12 (a) (i)	$S_n = n^2 + \frac{1}{3^n}$ $S_{n-1} = (n-1)^2 + \frac{1}{3^{n-1}}$ $T_n = S_n - S_{n-1}$ $= \left(n^2 + \frac{1}{3^n}\right) - (n-1)^2 - \frac{1}{3^{n-1}}$ $= 2n - 1 - \frac{2}{3^n}$
(ii)	$7 - \frac{2}{3^4} + 9 - \frac{2}{3^5} + 11 - \frac{2}{3^6} + 13 - \frac{2}{3^7} + 15 - \frac{2}{3^8}$ $= T_4 + T_5 + \dots + T_8$ $= S_8 - S_3$ $= \left(8^2 + \frac{1}{3^8}\right) - \left(3^2 + \frac{1}{3^3}\right)$ $= \frac{360613}{6561}$
	$S_{10^{20}} = \left(10^{20}\right)^2 + \frac{1}{3^{10^{20}}}$ Since $n = 10^{20}$ is very large, $\left(\frac{1}{3}\right)^{10^{20}}$ is negligible when compared to $\left(10^{20}\right)^2$. Therefore $S_{10^{20}} \approx \left(10^{20}\right)^2$. $\sqrt{S_{10^{20}}} \approx 10^{20}$
12 (b) (i) (ii)	$x_1 = 1 + x_0 = 1 + 1 = 2$ $x_2 = 2 + x_1 = 2 + 2 = 4$ $x_3 = 2 + x_2 = 3 + 4 = 7$ $x_4 = 4 + x_3 = 4 + 7 = 11$ $x_0 = 0 + 1$ $x_1 = 1 + 1$ $x_2 = 3 + 1$ $x_3 = 6 + 1$ $x_4 = 10 + 1$ $S_1 = 1 = T_1$ $S_2 = 3$ $\Rightarrow T_2 = 3 - 1 = 2$ $S_3 = 6$ $\Rightarrow T_3 = S_3 - S_2 = 3$ Hence $\{T_n\}$ forms an A.P. with first term 1 and common difference 1.

(iii)	Therefore $S_n = \frac{n}{2}(n+1)$. Hence the conjecture for x_n is given by $x_n = \frac{n}{2}(n+1)+1$, where $n \in \mathbb{N}, n \geq 0$. Let $P(n)$ be "$x_n = \frac{n}{2}(n+1)+1, n \in \mathbb{N}, n \geq 0$," When $n = 0$ LHS = $x_0 = 1$ RHS = $0 + 1 = \text{LHS}$ Therefore $P(0)$ is true. Assume $P(k)$ is true, i.e. $x_k = \frac{k}{2}(k+1)+1$ for some $k \in \{\mathbb{N}^+ \cup 0\}$ Prove that $P(k+1)$ is true, i.e. $x_{k+1} = \frac{k+1}{2}(k+2)+1, k \in \{\mathbb{N}^+ \cup 0\}$ LHS = x_{k+1} $= (k+1) + x_k$ $= (k+1) + \left[\frac{k}{2}(k+1)+1 \right]$ $= (k+1) \left[1 + \frac{k}{2} \right] + 1$ $= \frac{1}{2}(k+1)(k+2)+1$ = RHS Since $P(0)$ is true and $P(k)$ is true implies $P(k+1)$ is true, hence by mathematical induction, $P(n)$ is for true for all $n \in \{\mathbb{N}^+ \cup 0\}$.
-------	---

The End